

Plasmodium falciparum, epigenetic plasticity & Artemisinin Resistance: A classic case of Drug Failure

Atin Thakur

BS-MS Student, Indian Institute of Science Education and Research (IISER)

Note: This is a perspective article that synthesizes selected studies of *P. falciparum* adaptation and antimalarial resistance to highlight its epigenetic perspective.

Abstract

Malaria caused by *Plasmodium falciparum* poses a global health threat, particularly due to resistance against artemisinin-based combination therapies (ACTs). Resistance is traditionally viewed as a result of genetic mutations, with *PfKelch13* serving as a key molecular determinant. However, a more interesting question is how parasites generate adaptive phenotypic diversity prior to the emergence of stable resistance. Furthermore, the extent of nuclear–mitochondrial crosstalk in shaping stress adaptation remains largely unexplored. Together, these observations support a systems-level framework in which genome-wide epigenomic organization, enhancer-mediated regulation, histone modification networks, stress-responsive chromatin regulators, transcriptome-level variation, cellular heterogeneity, higher-order chromatin organization, and genome compartmentalization collectively drive adaptive plasticity, ultimately contributing to phenotypic adaptation and stable antimalarial resistance.

Keywords: *Plasmodium falciparum*, epigenetics, artemisinin resistance, PfGCN5, PfHDAC1, histone methyltransferases, cellular heterogeneity, stress adaptation, chromatin architecture

1. Introduction

Malaria remains one of the world's most significant infectious diseases. Despite global malaria elimination programs, the rising anti-malarial drug resistance remains a greater concern. Artemisinin-based combination therapies (ACTs) are the frontline treatment; however, delayed parasite clearance and resistance phenotypes are observed across multiple endemic regions. Initially, resistance was seen as a consequence of mere genetic mutations. While mutations in *PfKelch13* remain important molecular markers of artemisinin resistance, increasing resistance evidences reflect the multifactorial epigenetic modulations.

The aim of this study is to see how malaria epigenetics have transformed our understanding of parasite adaptation and resistance.

2. Epigenetic Organization of the *P. falciparum* Genome

2.1 Histone Modifications and Chromatin States

Genome-wide epigenomic analyses has shown that chromatin structure plays a central role in regulating parasite gene expression. Histone modifications define active, repressed, and poised transcriptional states. The poised chromatin is of prime importance as many genes remain transcriptionally prepared for activation during stress, developmental transitions, or host adaptation.

2.2 Histone Methylation and SET Domain Proteins

Unlike histone acetylation, which often mediates rapid transcriptional responses to environmental stimuli, histone methylation establishes long-lived chromatin states associated with activation, repression, or developmental surety. PfSET proteins catalyse the deposition of these methylation marks and thereby influence virulence gene expression, sexual commitment, heterochromatin formation, and parasite development. For example, methylation of H3K9 promotes recruitment of PfHP1, leading to chromatin compaction and silencing of subtelomeric *var* genes involved in antigenic variation. Similarly, methylation-dependent regulation of loci such as *PfAP2-G* regulates transition from asexual proliferation to gametocytogenesis. However, the increasing evidence linking chromatin regulation to artemisinin tolerance, I am unsure whether PfSET proteins function primarily as developmental regulators or as contributors to stress-adaptive phenotypes that eventually changes the chromatin state and might facilitate stable drug resistance.

3. Enhancer-Mediated Gene Regulation

The 462 enhancer-like element regions are enriched for H3K4me1, exhibit accessible chromatin, produce bidirectional transcripts, generate enhancer-associated RNAs, all hallmarks of active regulatory elements. Their gene regulation in malaria parasites seems beyond promoter control and involves distal regulatory interactions. As *P. falciparum* possesses a relatively limited repertoire of ApiAP2 transcription factors compared with the complexity of its developmental transitions and stress responses. Enhancers may expand the regulatory capacity of a limited ApiAP2 repertoire, allowing context-dependent transcriptional responses.

4. The Role of Histone Acetylation in Parasite Adaptation

4.1 Histone Acetyltransferases

Histone acetyltransferases (HATs) promote transcriptional activation by acetylating histone tails and increasing chromatin accessibility. Among them, PfGCN5 is a central regulator of stress-responsive transcription in *P. falciparum*. It preferentially occupies poised genes and loci involved in protein homeostasis, oxidative stress responses, and artemisinin tolerance under stress conditions. Inhibition of PfGCN5 sensitizes resistant parasites to artemisinin during drug-induced stress, suggesting adaptive transcriptional reprogramming.

4.2 Histone Deacetylases

Histone deacetylases (HDACs) counteract histone acetyltransferase activity by removing acetyl groups from histones, thereby modulating chromatin accessibility and gene expression. PfHDAC1 regulates genes associated with asexual proliferation, DNA replication, erythrocyte invasion, and developmental progression. PfHDAC1 overexpression enhances parasite survival following dihydroartemisinin exposure and is accompanied by transcriptional changes linked to stress adaptation. Furthermore, drug treatment alters the chromatin occupancy of PfHDAC1, redirecting it towards loci implicated in autophagy, DNA damage repair, and survival pathways.

“Whether PfHDAC1 directly mediates resistance or indirectly enhances fitness remains an important question”.

5. Cellular Heterogeneity as a Survival Strategy

Individual *P. falciparum* parasites exhibit distinct expression profiles despite sharing the same genetic background and developmental stage. Subpopulations characterized by elevated expression of stress-response genes, protein-folding chaperones, and metabolic adaptation pathways coexist within the population. Exposure to temperature stress further amplifies this heterogeneity, leading to the emergence of transcriptional states not observed under normal growth conditions. This might suggest that environmental stress does not create adaptive phenotypes de novo; rather, it selectively enriches pre-existing cellular states already present within genetically similar parasite populations.

6. Transcriptome-Level Variation and Phenotypic Diversity

High levels of RNA-only sequence variation are largely uncoupled from genomic polymorphisms. A substantial proportion of these variants result in amino acid substitutions implying that genetically identical parasites may produce distinct protein isoforms from the same genomic template. As a consequence, functional diversity might have arisen at the transcriptomic level without requiring permanent genetic change which explains the altered physiological states while preserving genomic integrity.

7. Higher-order chromatin organization and Virulence Regulation

The mutually exclusive expression of *var* genes is essential for antigenic variation and immune evasion in *P. falciparum*. Recent evidence demonstrates that PfHP1 undergoes phase separation to form dynamic heterochromatic condensates that compartmentalize virulence genes into transcriptionally repressive domains. PfHP1 preferentially associates with H3K9me3-marked chromatin, compacts AT-rich DNA, and promotes clustering of heterochromatic loci across chromosomes. Disruption of PfHP1 phase separation abolishes heterochromatin compartmentalization, weakens inter- and intrachromosomal contacts, and results in widespread derepression of *var* genes and other clonally variant gene families. These findings suggest that virulence regulation appears to depend not only on local gene control but also on the physical compartmentalization of chromatin domains within the nucleus.

8. Population Genomics and Multifactorial Resistance

The delayed parasite clearance despite the absence of *PfKelch13* resistance mutations, suggests alternative adaptive mechanisms which lead to reduced drug susceptibility. Variants affecting ubiquitination pathways, membrane transport systems, and organellar functions, including those associated with the mitochondrion, apicoplast, and digestive vacuole might have led to:

Stress → Epigenetic regulation → Phenotypic diversity → Selection of advantageous states → Possible genetic fixation.

9. WHAT I GOT CURIOUS ABOUT

1. Whether organelle-to-nucleus signaling contributes to the epigenetic reprogramming underlying adaptive phenotypes?
2. Are chromatin regulators direct mediators of resistance or general stress regulators?
3. What molecular mechanisms generate transcriptional heterogeneity?
4. How do enhancer RNAs regulate parasite gene expression?
5. What roles do SET-domain proteins play in quiescence?
6. Can epigenetic regulators be used to treat malaria?
7. If genetically identical parasites can occupy fundamentally different physiological states, what does it truly mean to call resistance a genetic trait?

References

1. **"PfHDAC1 is an essential regulator of *P. falciparum* asexual ..."** (2024) — Published in *eLife* / PMC [11](#)
2. **"A comprehensive epigenome map of *Plasmodium falciparum* reveals unique mechanisms of transcriptional regulation and identifies H3K36me2 as a global mark"** (2015) — *Epigenetics & Chromatin* [86](#)

3. **"Histone acetyltransferase PfGCN5 regulates stress responsive and artemisinin resistance related genes in Plasmodium falciparum"** (2021) — *Nature Scientific Reports* [2123](#)
4. **"Genome-wide identification of novel intergenic enhancer-like elements: implications in the regulation of transcription in Plasmodium falciparum"** (2017) — *BMC Genomics* [10](#)
5. **"Genome-wide survey and phylogenetic analysis of histone acetyltransferases and histone deacetylases of Plasmodium falciparum"** (2018) — *The FEBS Journal* [2619](#)
6. **"Genomic analysis of Indian isolates of Plasmodium falciparum"** (2023) — *Infection, Genetics and Evolution* [13](#)
7. **"Pervasive sequence-level variation in the transcriptome of Plasmodium falciparum"** (2022) — *NAR Genomics and Bioinformatics* [17](#)
8. **"The many paths to artemisinin resistance in Plasmodium falciparum"** (2023) — *Trends in Parasitology* (review) [25](#)
9. **"Phase separation of heterochromatin protein PfHP1 drives virulence gene regulation in Plasmodium falciparum"** (2025/2026) — *Cell Reports* [1224](#)
10. **"Yet to SET: Plasmodium falciparum Histone Lysine Methyltransferases"** (2025) — *ACS Infectious Diseases* [20](#)
11. **"Single-cell RNA sequencing reveals cellular heterogeneity and stage transition during Plasmodium falciparum development"** (2023, co-author) — referenced in *eLife* [7](#)